

Evolutionary and functional analyses reveal a role for the RHIM in tuning RIPK3 activity across vertebrates

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This **important** study provides **compelling** evidence for the evolutionary diversification and conserved NF κ B-inducing function of RHIM-containing RIP kinase proteins across animal lineages, combining thorough bioinformatic analysis with functional assays in human cells. The findings are of broad interest to immunologists and evolutionary biologists, though some novel observations would benefit from deeper conceptual integration.

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Abstract

Receptor interacting protein kinases (RIPK) RIPK1 and RIPK3 play important roles in diverse innate immune pathways. Despite this, some RIPK1/3-associated proteins are absent in specific vertebrate lineages, suggesting that some RIPK1/3 functions are conserved while others are more evolutionarily labile. Here, we perform comparative evolutionary analyses of RIPK1-5 and associated proteins in vertebrates to identify lineage-specific rapid evolution of RIPK3 and RIPK1 and recurrent loss of RIPK3-associated proteins. Despite this, diverse vertebrate RIPK3 proteins are able to activate NF- κ B and cell death in human cells. Additional analyses revealed a striking conservation of the RIP homotypic interaction motif (RHIM) in RIPK3, as well as other human RHIM-containing proteins. Interestingly, diversity in the RIPK3 RHIM can tune activation of NF- κ B while retaining the ability to activate cell death. Altogether, these data suggest that NF- κ B activation is a core, conserved function of RIPK3, and the RHIM can tailor RIPK3 function to specific needs within and between species.

Introduction

Receptor interacting protein kinase 3 (RIPK3), and the closely related kinase RIPK1, have critical roles in mediating cell death and inflammatory signaling [1 

1 (TNFR1), RIPK3 and RIPK1 interact via their shared RIP homotypic interaction motif (RHIM) to form amyloid-like aggregates [2], leading to their autophosphorylation and activation of additional effector proteins to carry out their innate immune functions. For example, RIPK3 can phosphorylate MLKL to activate highly inflammatory programmed necrotic cell death (necroptosis) [3], RIPK1/3 can engage caspase-8 (CASP8) to activate apoptotic caspases [4, 5], or RIPK1/3 can activate the pro-survival and pro-inflammatory transcription factor NF- κ B [6, 7]. Several factors are known to impact the outcome of RIPK1/3 activation, such as the activating signal and caspase activity [1, 8]. Altogether, RIPK3 and RIPK1 are central to determining cell fate downstream of various innate immune stimuli.

Like many innate immune proteins, RIPK3 is evolutionarily dynamic in vertebrates. Phylogenomic analyses of a small number of species from different vertebrate clades suggest that RIPK3, and the associated proteins ZBP1 and MLKL, have been lost in some vertebrate lineages including birds, carnivores, and marsupials [9, 10]. In addition, RIPK3, RIPK1, MLKL, and ZBP1 are known to interact with viral proteins [11–16] and have been shown to be evolving under recurrent positive selection in both primates and bats [17, 18], a hallmark of host proteins that directly interact with viral proteins [19]. Despite the known evolutionary divergence of RIPK3 in vertebrates, most of what is known about their activation and regulation is from studies conducted in humans and mice. Even between humans and mice, there are known differences in the roles of specific genes in regulation of NF- κ B, suggesting this pathway can be tailored to the distinct environments of different species [20]. These differences likely extend beyond humans and mice, and likely include RIPK1/RIPK3 mediated activation of NF- κ B. Furthermore, RIPK1 and RIPK3 are part of a larger family of RIPKs, including RIPK2, RIPK4, and ANKK1/RIPK5, which, with the exception of RIPK5, are known to activate NF- κ B in various cellular contexts [21–23]. The degree to which these other RIP kinases and their interacting partners have diversified across vertebrates is unknown.

Here, we use evolutionary and functional approaches to characterize diversification of RIPK3-mediated activation of NF- κ B. We identified RIPK1- and RHIM-mediated activation of NF- κ B by diverse vertebrate RIPK3 proteins, as well as distinct mechanisms of RIPK3-mediated activation of NF- κ B. Phylogenetic and phylogenomic analyses of RIPK1-5 and several key associated proteins revealed the dynamic evolution of RIPK3 and RIPK1. In addition to observing the loss of RIPK3 and necroptosis-associated proteins in some vertebrate lineages, we identify changes in regulatory features—including catalytic sites and CASP cleavage sites—in RIPK3 and RIPK1 in specific mammalian lineages that may affect their function. Intriguingly, we found that the RHIM domain of RIPK3 and RIPK1, as well as other RHIM-containing proteins, is highly conserved in vertebrates and in a non-vertebrate RIPK1 protein. Consistent with this strong conservation of the RHIM domain, we found that RIPK1 proteins from diverse vertebrates, and even a non-vertebrate RIPK1, are able to activate NF- κ B, suggesting that activation of this inflammatory response is a conserved function of RIPK3, RIPK1, and potentially other RHIM-containing proteins in vertebrates. Altogether, these data suggest that activation of NF- κ B is the ancestral and conserved feature of RIPK3 and RIPK1, while other functions such as necroptosis are more evolutionarily labile and rapidly evolving, likely as a response to evolutionary pressure from pathogens.

Results

Comparative evolutionary analysis of RIPK1-5

To understand the evolution and diversification of RIPK function, we first wished to compare the functions of human RIPKs. There are canonically four RIPKs in humans, RIPK1–4. ANKK1 is a close paralog of RIPK4 and has therefore been denoted as RIPK5. Additional proteins have also been described as RIPK proteins, including DSTYK, LRRK1, and LRRK2, but these proteins are phylogenetically distinct and will therefore not be considered here [23]. The shared domain

architecture of RIPK1-5 includes a conserved N-terminal kinase domain, an extended disordered intermediate domain, and distinct protein interaction domains at the C-terminus that can include a death domain (DD), RIP homotypic interaction motif (RHIM), caspase activation and recruitment domain (CARD), or ankyrin repeats (ANK) (**Figure 1A**). Human RIPK1-4 activate NF- κ B, albeit through different mechanisms, including differential dependence on endogenous RIPK1 and kinase activity (Supplementary Figure 1-3). Interestingly, despite its close homology to RIPK4, RIPK5 did not activate NF- κ B. Altogether, these data highlight NF- κ B as a shared function of RIPK1-4.

We next wished to perform comparative evolutionary analyses of RIPK1-5 in vertebrates to determine which RIPKs might be undergoing functional divergence. We first analyzed RIPK1-5 in four distinct mammalian clades to determine whether there was evidence of recurrent positive selection, which can be a hallmark of host-pathogen interactions and can drive functional diversification. Notably, we only found evidence for positive selection acting on RIPK1 and RIPK3 (**Fig 1B**, Supplementary files 1-2). Interestingly, RIPK1 and RIPK3 are not uniformly evolving under positive selection in all mammalian clades analyzed. For instance, in primates, RIPK3 is evolving under strong positive selection, where RIPK1 is not, consistent with previous data [17, 18]. In contrast, ungulate RIPK3 does not show evidence for positive selection whereas ungulate RIPK1 does. These data indicate that among RIPK1-5, only RIPK1 and RIPK3 genes have undergone recurrent positive selection, and that RIPK1 and RIPK3 may have faced different selective pressures across distinct mammalian lineages.

In addition to evolving under positive selection in primates and bats [17], RIPK3, ZBP1, and MLKL are known to be lost in some vertebrate lineages [9]. However, it is unknown if other RIPK and RIPK-associated proteins are similarly lost. We therefore performed phylogenomic analysis on 489 vertebrate species for the presence or absence of RIPK1-5 and several key RIPK-associated proteins (**Figure 1C**, Supplementary Files 4 and 5). Due to the difficulty in discerning whether a gene absence in an individual species is due to true gene loss or incomplete genome assembly, we focused on multi-species patterns of gene loss within and between vertebrate lineages. Given that NF- κ B is broadly present in many vertebrate and non-vertebrate species, including horseshoe crab [24], *Drosophila melanogaster* [25], and cnidarians and bivalves (reviewed in [26]), we analyzed proteins associated with other RIPK functions. We first confirmed the loss of necroptosis associated proteins that has been previously reported [9]: RIPK3 and ZBP1 in birds; RIPK3, ZBP1, and MLKL in marsupials; MLKL in carnivores. In addition, we found that ZBP1 is absent in all fish species, suggesting that this protein, and ZBP1-mediated necroptosis, arose only during tetrapod evolution.

We then analyzed RIPK1 and its associated proteins. RIPK1 is found nearly every vertebrate genome queried, as well as clear homologs in several non-vertebrate species, including zebra mussel (*Dreissena polymorpha*), several additional spiralia species, and lancelet species in the genus *Branchiostoma* (accession numbers in Supplementary File 4). Homologs—defined as proteins with a homologous N-terminal kinase domain and a C-terminal RHIM tetrad and death domain—were not found in tunicates, echinoderms, cnidarians, nematodes, arthropods, or amoeba. Additional RIPK-like proteins have been found in other non-vertebrate species, including a RIPK1/2-like protein in *Drosophila* and other protostomes [9], although these proteins were not considered here due to minimal sequence similarity to the human RIP kinase domains and our focus on vertebrate RIP kinases. We also identified caspase-8 (CASP8), a known regulator of RIPK1, and its closest homolog, CASP10. Due to the similar domain structure of CASP8 and CASP10, we only identified whether a species has a single CASP8/10 protein, two distinct CASP8/10 proteins as determined by phylogeny (see Materials and Methods), or none. At least one copy of CASP8/10 is found in most tetrapod genomes, although it is absent sea lamprey. Two copies, presumably marking the delineation between CASP8 and CASP10, began in tetrapod, coinciding with the emergence of ZBP1. This may reflect the role of CASP8 in regulating ZBP1, and duplication of this caspase allowed for the specialized function of CASP8 [27]. We also confirmed the loss of

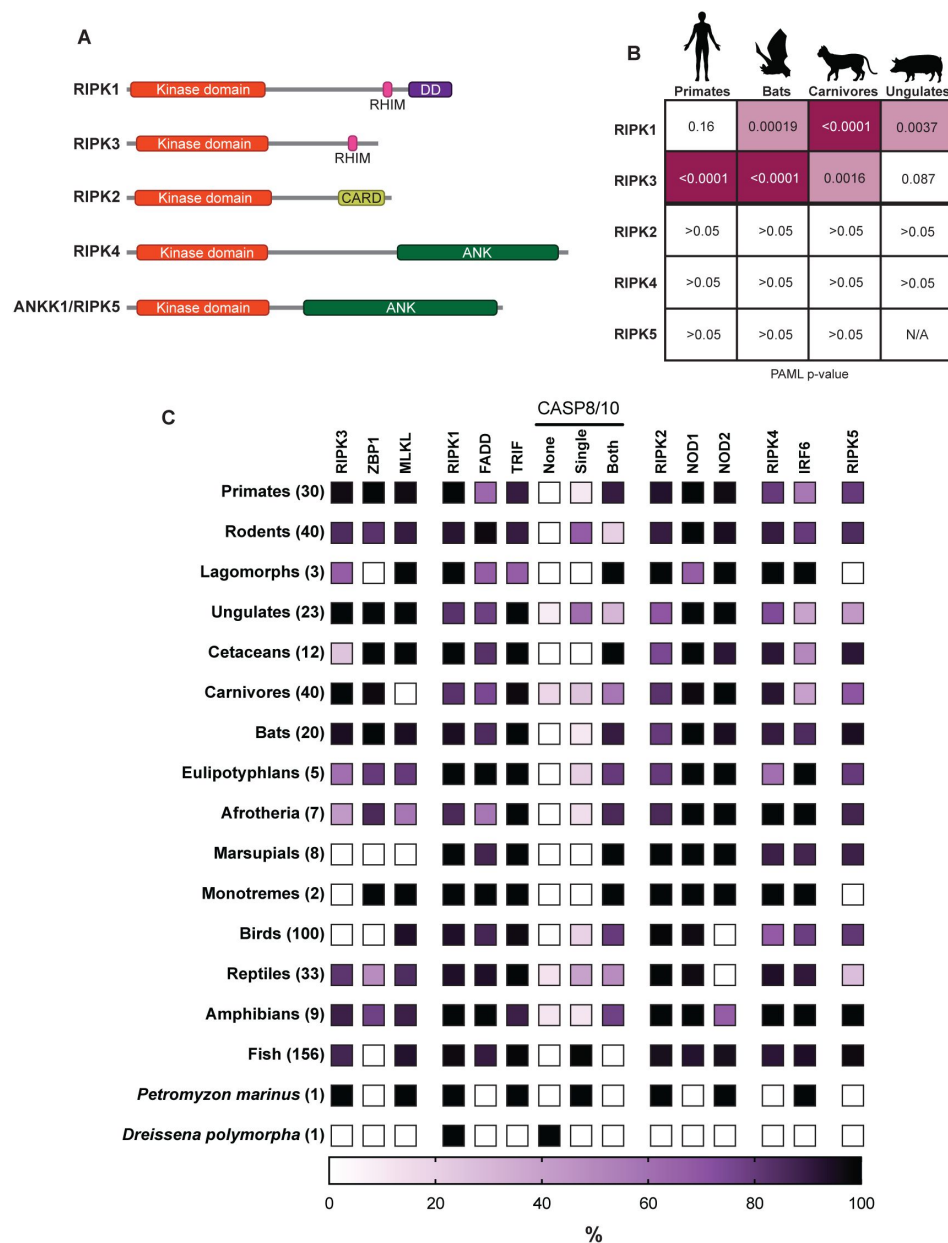


Figure 1.

Comparative evolutionary analysis of RIPK1-5.

(A) Domain structures of human RIP kinases. RHIM = RIP homotypic interaction motif, DD = death domain, CARD = caspase activation and recruitment domain, ANK = ankyrin repeats. (B) Positive selection analysis of RIPK1-5 in the indicated mammalian order. Input sequences and PAML p-values can be found in Supplementary Files 1 and 2. (C) Heat map showing the percentage of species within a clade that have the indicated protein. The clades and the number of species within each clade are indicated on the left. Complete lists of proteins and species in each group can be found in Supplementary Files 4 and 5. Lancelet species were not queried for all proteins and are not included in this graph.

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CASP10 in some rodents [28] (see Supplementary Figure 4A for CASP8/10 breakdown), revealing that suborder Myomorpha (“mouse-like”) and Castorimorpha (“beaver-like”) rodents have lost CASP10, whereas Sciuromorpha (“squirrel-like”) and Hystricomorpha (“porcupine-like”) rodents have retained CASP10. Finally, we found that FADD and TRIF are well conserved in all vertebrates.

We also analyzed the phylogenomic distribution of RIPK2, 4, and 5 and their associated proteins. RIPK2 and its associated protein NOD1 are found in most vertebrates [21, 22]. Intriguingly, we observed a second paralog of RIPK2, which we call RIPK2B, in many vertebrates but lost in therian (live-bearing) mammals (Supplementary Figure 4B). NOD2 is present in lamprey and fish, but absent in birds, reptiles, and some amphibians, including some of the same species that contain RIPK2B. While RIPK2B retains both the kinase domain and CARD, there may be distinct regulatory mechanisms, specifically those related to NOD2, compared to RIPK2. RIPK4 is present in jawed vertebrates. IRF6, the only known phosphorylation target of RIPK4 [21, 22], emerged prior to RIPK4 and has been lost in many vertebrate species, particularly in carnivores, cetaceans, and ungulates. RIPK5, which has no known function and no known interacting proteins, has been lost in several lineages, including monotremes and lagomorphs. Overall, our analyses revealed that RIPK1 and RIPK3 are evolutionarily distinct from other RIP kinases, and the recurrent loss of some RIPK1/3-associated necroptosis proteins (ZBP1, MLKL, CASP8) and RIPK3 suggests that these proteins may have functions tailored to the needs of specific vertebrate lineages.

Diverse vertebrate RIPK3 proteins activate NF-κB

Our analyses revealed dynamic evolution of RIPK3 in vertebrates, including signatures of positive selection in multiple mammalian clades and recurrent loss of RIPK3 and its associated proteins. We hypothesized that this rapid RIPK3 evolution may have resulted in different functions of RIPK3 in different vertebrate lineages. We therefore tested the ability of RIPK3 proteins from diverse vertebrates, ranging from 23–63% similarity to humans (Fig 2A, Supplementary Figure 5), to activate NF-κB in human cells. We included two mammalian species that retain all necroptosis proteins (mouse, *Mus musculus*; pig, *Sus scrofa*), a mammalian species lacking a necroptosis protein (cat, *Felis catus*), reptilian species that do (turtle, *Chelonia mydas*) and do not (lizard, *Anolis carolinensis*) retain all necroptosis proteins, and sea lamprey (*Petromyzon marinus*) as our most distal vertebrates relative to humans. To our surprise, among all vertebrates tested, only pig and sea lamprey RIPK3 failed to activate NF-κB in human cells (Fig 2B). This striking conservation of function indicates that NF-κB activation has remained intact despite lineage specific RIPK3 divergence and loss of interacting partners. We then analyzed whether, like human RIPK3 (Supplementary Figure 1C) [29], other vertebrate RIPK3 proteins require RIPK1 to activate NF-κB by repeating experiments in RIPK1 KO cells. Interestingly, while mammalian RIPK3 activation is dependent on RIPK1, both lizard and turtle RIPK3 activate NF-κB independently of RIPK1 (Fig 2B). These results indicate that the requirements for NF-κB activation by RIPK3 are species-specific.

Finally, we tested whether another function of RIPK3, activation of ZBP1- and MLKL-dependent cell death (Supplementary Figure 6), is also conserved. Most mammalian RIPK3 proteins were able to activate cell death using human ZBP1 and MLKL (Fig 2C). This included pig RIPK3, which is unable to activate NF-κB, suggesting that activation of cell death and NF-κB by RIPK3 proteins have been separated in a species-specific manner. Likewise, we observed that cat RIPK3 can activate NF-κB but fails to activate cell death. Based on this observation, we analyzed the sequences of carnivore RIPK3 proteins and found that feline RIPK3 have mutated kinase catalytic residue (Supplementary Figure 7), explaining the loss of cell death activation by cat RIPK3. Outside of mammals, only lizard RIPK3 was able to activate cell death, despite the ability of both lizard and turtle RIPK3 to activate NF-κB, further suggesting that these functions of RIPK3 have distinct, species-specific requirements.

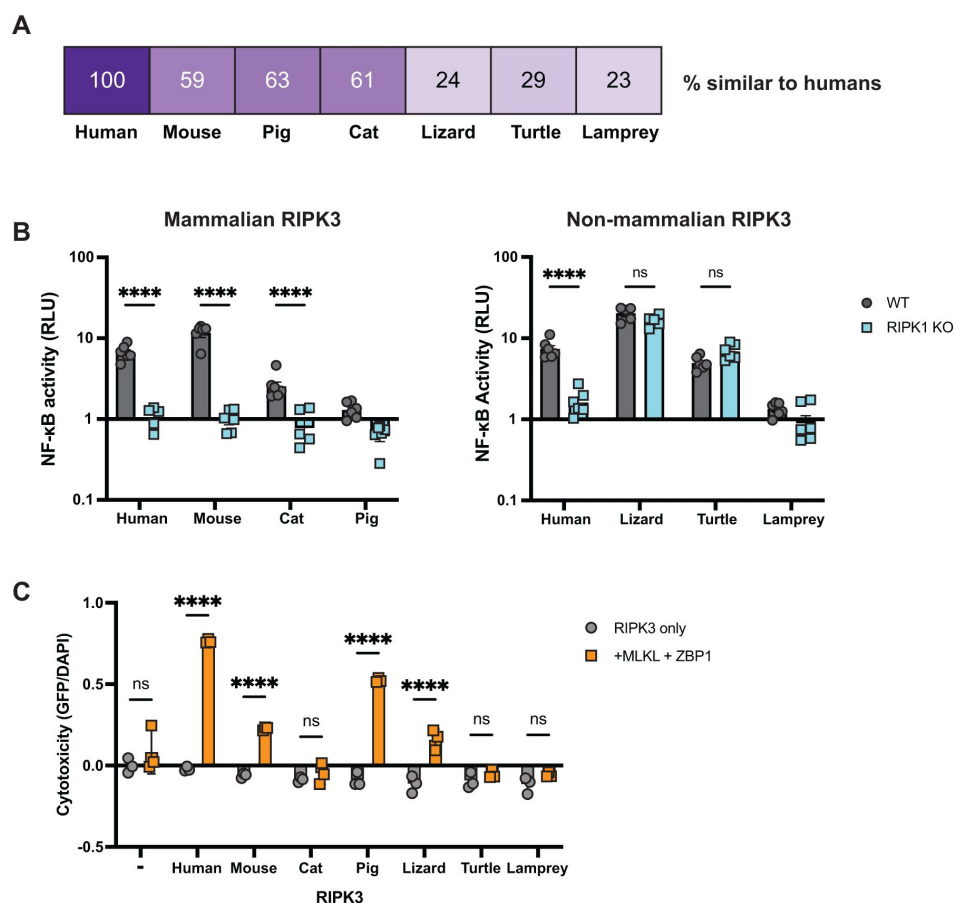


Figure 2.

Diverse vertebrate RIPK3 proteins activate NF-κB.

(A) Percent similarity of RIPK3 from the indicated species compared to humans. (B) RIPK3 proteins were transfected into WT or RIPK1 KO HEK293T cells, along with NF-κB firefly luciferase and control renilla luciferase reporter plasmids (see Materials and Methods), and NF-κB activity was measured at 18h post-transfection. (C) RIPK3 proteins were transfected into HEK293T cells with and without human ZBP1 and MLKL. At 18h post-transfection, cells were stained using the ReadyProbe Cell Viability kit and fluorescence was measured using a plate reader. Species shown are mouse (*Mus musculus*), cat (*Felis catus*), pig (*Sus scrofa*), lizard (*Anolis carolinensis*), turtle (*Chelonia mydas*), and lamprey (*Petromyzon marinus*). Data are representative of 3-5 independent experiments with n=3-6 replicates per group. Data were analyzed using two-way ANOVA with Šidák's multiple comparisons test. ns = not significant, **** = $p < 0.0001$.

Conservation of the RHIM sequence determines RIPK3 activation of NF- κ B

The conservation of NF- κ B activation by RIPK3 proteins that share as little as 24% sequence identity and are up to >300 million years diverged [30] suggests some regions of the protein are likely highly conserved. To identify such regions, we returned to our positive selection analyses. While we identified rapidly evolving sites throughout the primate RIPK3, there was a cluster of sites in and around the C-terminal RHIM (Fig 3A Supplementary File 5), which is a known region of interaction between RIPK1 and RIPK3 and RIPK3 homooligomerization [2, 31]. Intriguingly, despite multiple rapidly evolving sites within the RHIM, a core tetrad (VQVG) is not rapidly evolving. Similarly, the core tetrads of bat and carnivore RIPK3 are not rapidly evolving (Fig 3B). Furthermore, the RIPK3 core RHIM tetrad is highly conserved across vertebrates (Fig 3C). This is despite loss of other functional motifs in specific lineages, including loss of the RIPK3 catalytic site in felines (Supplementary Figure 7).

We therefore tested whether this conserved RHIM tetrad was responsible for activation of NF- κ B by diverse vertebrate RIPK3s. We mutated the RIPK3 RHIM core tetrad (V/I-Q-V/I-G to AAAA) in our panel of vertebrate RIPK3 proteins and tested the ability of these proteins to activate NF- κ B in human cells. Mutation of human, mouse, and cat RIPK3 RHIMs reduced activation of NF- κ B as expected (Fig 3D, Supplementary Figure 5). Surprisingly, despite activating independently of RIPK1 (Fig 2B), activation of NF- κ B by lizard and turtle RIPK3 remains dependent on the RHIM (Fig 3E, left), and independent of RIPK3 kinase activity (Supplementary Figure 8-9). These data support that reptile RIPK3 activates NF- κ B through a distinct mechanism from mammalian RIPK3, but one that is still highly dependent on the RHIM core tetrad. Interestingly, while analyzing RHIM domains of non-human RIPK3, we observed that the sea lamprey RHIM diverges from the canonical V/I-Q-V/I-G tetrad and instead contains TQIG (Fig 3C). To determine if this was responsible for the limited activation of NF- κ B by sea lamprey RIPK3, we mutated this motif to IQIG and observed an increase in NF- κ B activation (Fig 3E, right). Altogether, these data indicate that NF- κ B activation is a conserved function of RIPK3 that requires conservation of the core RHIM tetrad, but allows for substantial sequence divergence elsewhere in the protein.

NF- κ B activation is a shared function of RHIM-containing proteins and can be tuned by the RHIM

Given that activation of NF- κ B by RIPK3 is dependent the RHIM, we hypothesized that the core RHIM tetrad and activation of NF- κ B would be broadly conserved across all RHIM-containing proteins, including RIPK1, ZBP1 and TRIF. We found that the RIPK1 RHIM is highly conserved in vertebrates and in RIPK1-like proteins found in zebra mussel, *Linghula* species, and multiple lancelet species (Fig 4A). Human RIPK1 is known to activate NF- κ B independently of its RHIM through polyubiquitination of the intermediate domain [32, 33]. However, the strong conservation of the RHIM domain tetrad sequence (V/I-Q-V/I-G) of RIPK1 across species led us to hypothesize that RHIM-mediated activation of NF- κ B by RIPK1 is also conserved. We therefore tested a diverse panel of metazoan RIPK1 proteins in our NF- κ B assay. To avoid the effects of both kinase activity and kinase-mediated interactions [34], we generated RIPK1 C-terminus (RIPK1^{CT}) proteins from diverse species that do (mouse, pig) and do not (cat, lizard, lamprey, zebra mussel) retain all necroptosis proteins (Supplementary Figure 10). Both WT and RHIM mutant human RIPK1^{CT} activate NF- κ B to the same extent as full length RIPK1 (Supplementary Figure 11), validating our use of this system to characterize RIPK1 functions independent from the kinase domain. Diverse vertebrate RIPK1^{CT} proteins activate NF- κ B independent of endogenous RIPK1 (Supplementary Figure 12), consistent with our hypothesis that NF- κ B activation is an ancestral and conserved function of these proteins. However, like RIPK3, the specific sequence requirements for RIPK1-mediated NF- κ B activation varied by species. Specifically, mutation of the RHIM domain does not affect activation of NF- κ B by mammalian proteins, including human,

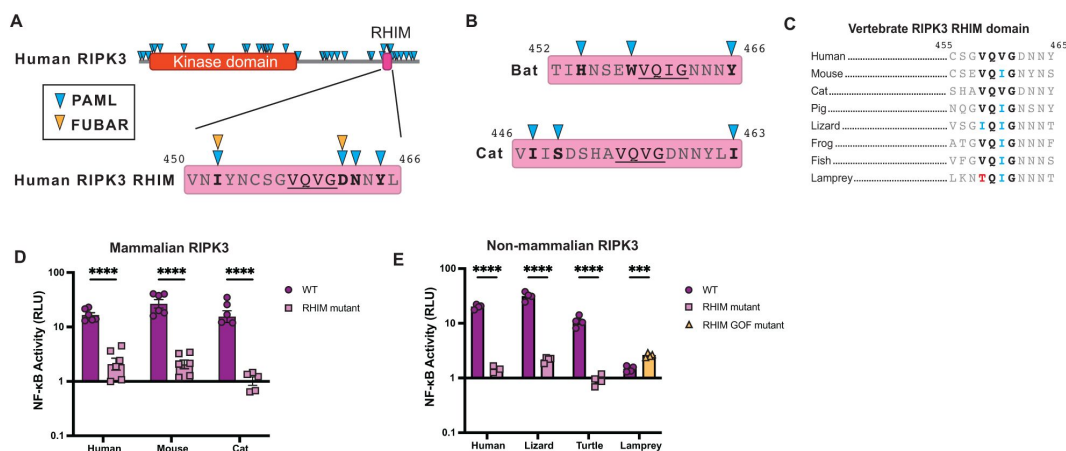


Figure 3.

Conservation of the RHIM sequence determines RIPK3 NF-κB activation.

(A) Residues evolving under positive selection as identified using PAML and FUBAR (see Materials and Methods) in primate RIPK3 and the primate RIPK3 RHIM domain mapped on the human sequence. (B) Residues evolving under positive selection in bat and carnivore RIPK3, mapped on to the *Sturnira hondurensis* and *Felis catus* sequences respectively. (C) Alignments of the RIPK3 RHIM across diverse vertebrates. Residue numbers refer to the human sequence. (D-E) Mammalian (D) and non-mammalian (E) RIPK3 proteins were transfected into WT HEK293T cells along with NF-κB firefly luciferase and control renilla luciferase reporter plasmids (see Materials and Methods). NF-κB activity was measured at 18h post-transfection. Species shown are mouse (*Mus musculus*), cat (*Felis catus*), pig (*Sus scrofa*), lizard (*Anolis carolinensis*), turtle (*Chelonia mydas*), and lamprey (*Petromyzon marinus*). Data are representative of 3-5 independent experiments with n=3-6 replicates per group. Data were analyzed using two-way ANOVA with Šidák's multiple comparisons test. ns = not significant, **** = p < 0.0001.

mouse, cat, and pig (**Figure 4B**, left). In contrast, mutation of the RHIM domain attenuated or completely prevented the ability of non-mammalian (lizard, sea lamprey, and zebra mussel) RIPK1^{CT} to activate NF- κ B (**Figure 4B**, right).

Other than RIPK1 and RIPK3, only two other human proteins contain a RHIM domain, TRIF and ZBP1. The TRIF RHIM and the first RHIM of ZBP1 are also highly conserved in mammals and some other tetrapods, whereas the second RHIM of ZBP1 diverges even within mammals (**Fig 4C**, Supplementary Figure 13A). We then tested for RHIM-dependence of NF- κ B activation by human TRIF and ZBP1. As described previously [35], ZBP1 activates NF- κ B in a RHIM- and RIPK1-dependent manner (**Fig 4D**). Consistent with our evolutionary prediction, only the first RHIM tetrad is required for NF- κ B activation by ZBP1 (**Fig 4D**). Conversely, mutation of the TRIF RHIM domain does not affect activation of NF- κ B, and activation is only moderately reduced in RIPK1 KO cells (Supplementary Figure 13B-C), likely due to the TIR domain in TRIF that is known to mediate its innate immune signaling [36].

Finally, due to the central role of the core tetrad of the RHIM domain, we tested the plasticity of RHIM function with regard to sequence divergence. In our phylogenetic analyses of RIPK3, we identified several species in which the RHIM core tetrad has diverged, particularly outside of tetrapods, including tetrads that have diverged from the conserved motif (V/I-Q-V/I-G) at only a single residue (Supplementary Table 2). To test for functional differences, we inserted these naturally-occurring RHIM tetrads into human RIPK3 and characterized the ability to activate NF- κ B and cell death. We included tetrads found in rodents, bats, eulipotyphlans, reptiles, and fish (VQFG), amphibians and fish (LQIG, VQSG), fish (CQIG), and sea lamprey (TQIG). Interestingly, while most tested variants activate NF- κ B compared to the inactive RIPK3 RHIM mutant (AAAA tetrad), albeit lower than the WT (VQVG) tetrad, TQIG and VQSG did not activate NF- κ B (**Fig 4E**, Supplementary Figure 14-15). These data suggest that, while there is some plasticity in the RHIM tetrad, not all residues are functional. All tetrad variants were able to activate ZBP1-dependent cell death similar to WT RIPK3 (**Fig 4F**), further revealing the separation of RIPK3-mediated NF- κ B activation from cell death activation. These differences may be due to differential amyloid structures formed during heterooligomerization of the RIPK1 RHIM and RIPK3 RHIM versus homooligomerization of RIPK3 [2]. Altogether, these data suggest that diversity in the RIPK3 RHIM domain may tune activation of NF- κ B and tailor RIPK3 function to the specific needs of species.

Discussion

Activation of NF- κ B is critical to both the innate and adaptive immune response in humans and mice [20], and is a deeply conserved pathway in many metazoans [24-26]. Despite this deep conservation of NF- κ B signaling, differences in the function and regulation of NF- κ B-associated proteins between mice and humans have been identified [20], suggesting that this functionally well-conserved pathway can be adapted to different species. These data indicate that broader phylogenetic analysis of the genes and functions that are associated with NF- κ B signaling are needed to understand immune responses across species, including those that harbor pathogens that pose a zoonotic threat to humans. Importantly, the approach used here can be applied to other NF- κ B-associated proteins, and to other innate immune pathways, to better define species-specific immune responses.

Here, we apply broad phylogenetic and functional sampling to the RIP kinase family of NF- κ B activators. Using a phylogenomic approach, we identified widespread conservation of RIPK1, RIPK2, and RIPK4 and their associated proteins within vertebrates. We also show conservation of activation of NF- κ B signaling by RIPK1 and RIPK3 across diverse vertebrate species that is largely reliant on the highly conserved core tetrad of the RHIM domains found in both proteins. This striking conservation of RIPK1- and RIPK3-mediated activation of NF- κ B, from species that span

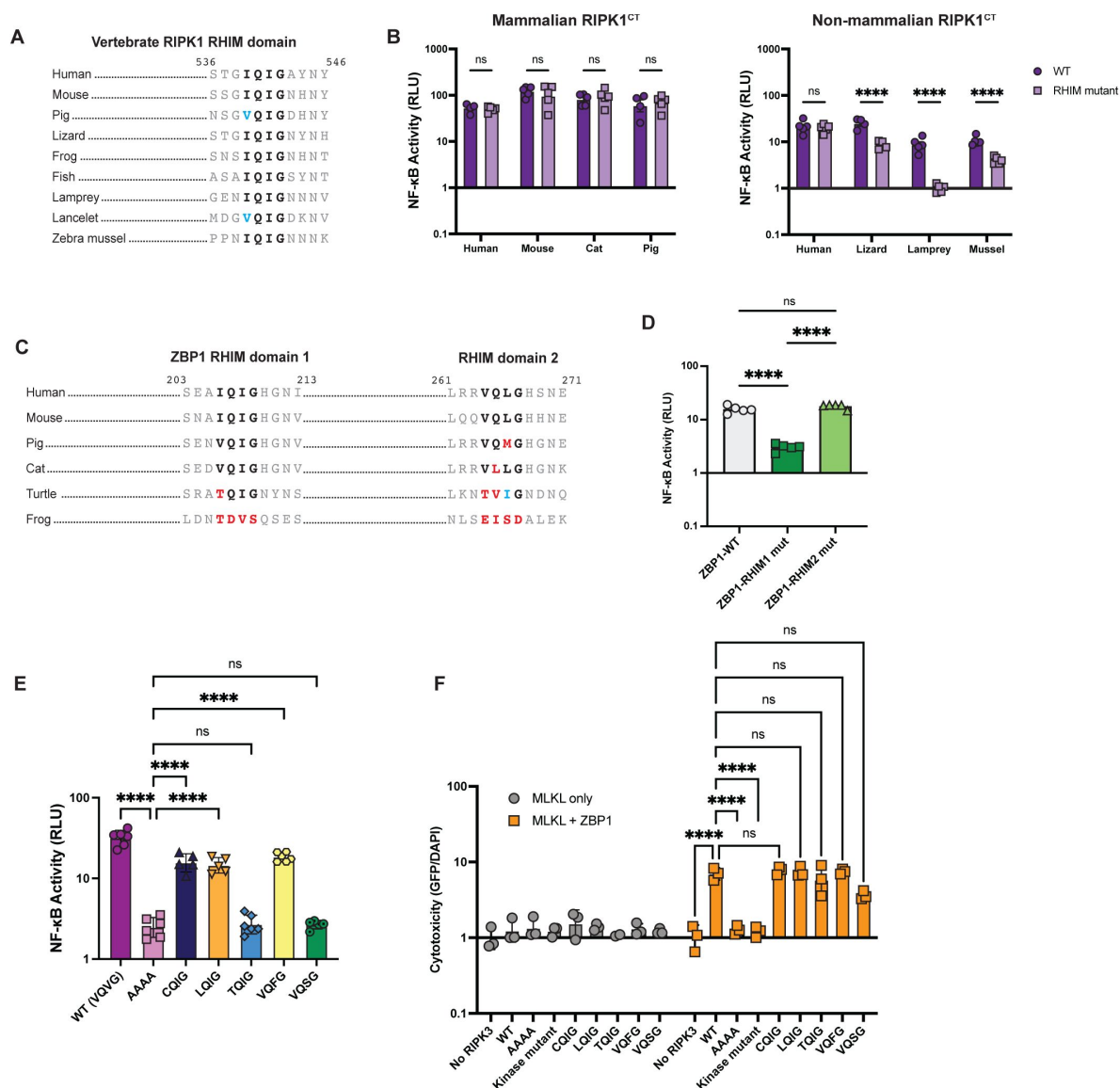


Figure 4.

NF-κB activation is a shared function of RHIM-containing proteins and can be tuned by the RHIM.

(A) Alignment of RIPK1 RHIM across diverse vertebrates. Residue numbers refer to the human sequence. Lancelet species is *Branchiostoma floridae*. (B) Diverse vertebrate RIPK1^{CT} proteins were transfected into HEK293T cells along with NF-κB firefly luciferase and control renilla luciferase reporter plasmids (see Materials and Methods), and NF-κB activity was measured at 18h post-transfection. (C) Alignment of ZBP1 RHIMs across diverse vertebrates. Residue numbers refer to the human sequence. (D) Activation of NF-κB by WT and RHIM mutant ZBP1 proteins. (E) NF-κB activation by human RIPK3 with the indicated RHIM tetrad variant. (F) Human RIPK3 proteins with the indicated RHIM tetrad variants were transfected into HEK293T cells with MLKL (gray circles) or MLKL and ZBP1 (orange squares) and viability was measured at 18h post-transfection. Data are representative of 2-5 independent experiments with n=3-6 replicates per group. Data were analyzed using two-way ANOVA with Šidák's multiple comparisons test (A, D) one-way ANOVA with Tukey's multiple comparisons test (E), or two-way ANOVA with Tukey's multiple comparisons test (F). ns = not significant, **** = p<0.0001.

>500 million years of vertebrate evolution, underscores the core functionality of RIP kinases as NF- κ B activating proteins. Despite this conservation of NF- κ B activation, we also observed lineage specific changes in RIPK presence or mechanisms of NF- κ B activation. For instance, we observed recurrent gene loss of RIPK3 and RIPK5 throughout the vertebrate phylogeny, and discovered a second paralog of RIPK2 in many non-mammalian vertebrate species (RIPK2B). In addition, although RIPK3-mediated activation of NF- κ B is highly dependent on RIPK1 across diverse mammals, RIPK3 activation of NF- κ B in reptiles is independent of RIPK1. Activation of NF- κ B is part of the antiviral immune response and is known to be antagonized by viruses across a range of species, including vertebrates and non-vertebrates [37–39]. This antagonism could result in adaptation of NF- κ B-associated proteins, including RIP kinases, leading to lineage-specific mechanisms of activation that are tailored to the specific contexts within and between species.

The most striking differences in RIPK-associated genes and functions are those involved in necroptosis. Necroptosis is highly inflammatory and the associated proteins must therefore be tightly regulated, as it could be a cause of immunopathology during infection. Despite this, necroptosis has been shown to be important for the response to several viruses [40–42]. It is hypothesized that necroptosis, defined by RIPK3-mediated activation of MLKL, arose in vertebrates [9, 43]. Interestingly, we find that two critical regulators of necroptosis, CASP8 and ZBP1, are only found in tetrapods, suggestive of an additional innovation within the necroptosis pathway during the divergence of tetrapods. Moreover, consistent with previous observations [9, 10], we observe a large number of necroptosis-associated gene losses, while also discovering important functional motifs in necroptosis proteins in distinct tetrapod lineages. These data support a model where necroptosis has been constructed as an important facet of the innate immune response in some tetrapods, but is not universal and has been deconstructed in species where the immunopathologic effects outweigh the benefits.

There have been several hypotheses to explain the loss of necroptosis associated proteins in specific vertebrate lineages, such as diet (*e.g.*, carnivores) or behavior (*e.g.*, torpor in reptiles) [9]. While the cause of a specific instance of gene loss is difficult to identify, it is clear that gene loss can provide an evolutionary tool for adaptation [44]. For example, caspases, which are key regulators of both immunologically silent apoptosis and inflammatory pyroptosis, differ greatly across mammals, with lineage-specific caspase repertoires including both loss and gain of novel caspases [28]. Even within a single mammalian order, homologous caspases have been found to serve different functions in response to pathogens [45]. This type of lineage-specific innovation likely exists for necroptosis, and gene loss could be a cause or a result of adaptations. Indeed, differences in MLKL activation by RIPK3 across species have been identified [46], which may be the result of additional diverse regulatory mechanisms for necroptosis even amongst species with an intact pathway. Strikingly, despite the lack of a complete necroptosis pathway in some lineages (cat, turtle, lamprey), our functional data indicate that RIPK3 retains the ability to activate NF- κ B. Activation of NF- κ B could therefore represent the ancestral function of RIPK3, and it has been co-opted to activate necroptosis in vertebrates.

The RHIM core tetrad was previously known to be conserved between humans and mice [47]. Here, we expand this analysis and identify striking conservation of the RIPK3, RIPK1, ZBP1, and TRIF RHIM domains across most vertebrates. Additional RHIM-containing proteins have been identified in non-vertebrate animals and in fungi [48]. Interestingly, the RHIM proteins found in fruit fly and fungi are known to be involved in antibacterial defense and cell death, respectively, suggesting that RHIM-mediated interactions may be an ancient mechanism of innate immunity. While activation of NF- κ B by all tested non-human RIPK3 proteins was highly dependent on the RHIM domain, only non-mammalian RIPK1^{CT} proteins required the RHIM for maximal NF- κ B activation. These species (lizard, lamprey, and zebra mussel) also lack ZBP1, with lamprey and zebra mussel predating the emergence of ZBP1 in tetrapods. Necroptosis may have driven evolution of RHIM-independent activation of NF- κ B by RIPK1. This may have also shaped RHIM-dependent functions of RIPK3 across vertebrates, as the RHIM tetrad is much less conserved in fish

compared to tetrapods, and insertion of these tetrad variants into human RIPK3 alters its function. Virus antagonism has also likely played a role in the evolution of RIPK3 and RIPK1 function. For example, vaccinia virus E3 prevents Z-DNA sensing by ZBP1 [40], human cytomegalovirus UL36 promotes MLKL degradation, and poxviruses encode an MLKL homolog to prevent RIPK3-mediated activation of host MLKL [49]. Conservation of the RHIM domain may be a way to retain some RIPK3/RIPK1 innate immune functions in the face of such antagonism. However, several viruses have taken advantage of this conservation and encode RHIM-containing proteins to prevent RHIM-dependent functions [43, 47]. Intriguingly, several of these mechanisms are known to be species-specific and differ between humans and mice [14, 15]. These data support a model where viruses and host necroptosis are engaged in a molecular arms race that has shaped the function of RIPK3 and RIPK1 across vertebrate species.

Altogether, our combined phylogenetic and functional approaches reveal both conservation and a striking lack of conservation in vertebrate RIPKs and their associated proteins and functions. Such characteristics are common hallmarks of pathways that are required for the innate immune response to pathogens, but are under constant evolutionary pressure to innovate to avoid pathogen antagonism. The strong conservation of NF- κ B activation by RIPK1 across species is consistent with the model in which RIPK1 function is central to determining cell fate downstream of TNF and TLR signaling [50]. Moreover, because some pathogens have evolved strategies to inhibit CASP8, RIPK3- and RIPK1-mediated necroptosis is activated as a secondary cell death strategy when CASP8 is inhibited [51]. This places both RIPK3 and RIPK1, critical regulators of necroptosis, at the center of determining cell fate during pathogen infection. The divergence of RIPK3 and RIPK1 across vertebrates identified here, as well as the previous finding that these proteins are evolving under positive selection [10, 17, 18], reveals that pathogens are likely the driving force behind the lineage-specific regulatory mechanisms or functions of these proteins across species that we observe here. Thus, the functional divergence identified in our work highlights the evolutionary innovation that can arise in innate signaling pathways across diverse species, and underscores the importance of considering natural diversity when characterizing the innate immune response to pathogens. Our work provides a model for further study on the diversity of innate immune proteins, protein families, and pathways across vertebrate and non-vertebrate species.

Materials & methods

Positive selection analysis

Positive selection analysis was performed using three independent methods as we have done previously [52, 53]. Nucleotide sequences from the indicated mammalian clade that aligned to full-length human RIPK1-5 were downloaded from NCBI and aligned using ClustalOmega [54]. Maximum likelihood (ML) tests were performed with codeml in the PAML software suite [55]. For PAML, aligned sequences were subjected to ML tests using NS sites models disallowing (M7) or allowing (M8) positive selection. The reported p-values were calculated using a chi-squared test on twice the difference of the log likelihood (lnL) values between the two models using 2 degrees of freedom. We confirmed convergence of lnL values by performing each analysis using two starting omega (dN/dS) values (0.4 and 1.5). Codons evolving under positive selection from PAML analyses have a posterior probability greater than 0.90 using Native Empirical Bayes (NEB) and Bayes Empirical Bayes (BEB) analysis and the F61 codon frequency model. The same nucleotide alignments were used as input for FUBAR (Fast, Unconstrained Bayesian AppRoximation) [56] and MEME (Mixed Effects Model of Evolution) [57] using the DataMonkey server [58]. In both cases, default parameters were used and codons with a signature of positive selection with a p-value of <0.01 are reported. Accession numbers for input sequences, PAML p-values, and sites identified can be found in Supplementary Files 1-3.

Gene loss analysis

Gene loss analysis was completed by first building phylogenetic trees for each protein of interest. Phylogenetic trees were generated for RIPK1-5 using previously described methods [59]. Briefly, the kinase domains from human RIPK1 (NP_001341859.1), RIPK2 (NP_003812.1), RIPK3 (NP_006862.2), RIPK4 (NP_065690.2), and RIPK5 (NP_848605.1) were used to query the RefSeq protein database using BLASTp with an e-value cutoff of 2×10^{-26} [60]. While LRRK1 and LRRK2 share kinase homology with RIPK1-5 and are sometimes known as RIPK6 and -7 [23], they have a distinct domain structure and function compared to RIPK1-5 and will therefore not be considered as RIP kinases for the purposes of this study. The five resulting sequence lists were consolidated to a single list of unique sequences using Geneious software. These sequences were then aligned using Clustal Omega [54] with two refinement iterations implemented in Geneious software. Following alignment, we filtered sequences based on alignment of an N-terminal kinase domain to human RIPK kinase domains and the presence of a C-terminal domain. Proteins with additional N-terminal domains, proteins lacking homology to human kinase domains, and proteins lacking C-terminal domains were removed. Using the aligned and filtered vertebrate RIPK sequences (4,550 sequences in total), a tree was built using FastTree [61, 62] implemented using Geneious. Vertebrate MAPKK7 sequences were used as an outgroup to help isolate RIP kinases from similar, but distinct, kinases. Using this tree, we were able to identify and extract *bona fide* RIP kinases. The extracted sequences were again aligned using Clustal Omega and filtered using the criteria above. A second tree was generated using FastTree. From this tree, we extracted individual sequence lists for RIPK1-5 to determine which vertebrate species did or did not have each RIPK. This method was repeated for, MLKL, TRIF, FADD, CASP8 and 10, NOD 1 and 2, and IRF6. The list of species and lists of accession numbers for each protein can be found in Supplementary Files 4 and 5. A list of outgroups used for each analyzed protein can be found in Supplementary File 6.

Plasmids and constructs

Coding sequences for human RIPK1 (Addgene #78834), human RIPK2 (ORFeome ID #4886), human RIPK3 (Addgene #78804), mouse RIPK1 (Addgene #115341), and mouse RIPK3 (Addgene #78805) were cloned into pcDNA5/FRT/TO backbone (Invitrogen, Carlsbad, CA) with an N-terminal V5 tag and linker using Gibson Assembly (New England Biolabs, Ipswich, MA). Human RIPK4 (NP_065690.2) and RIPK5 (NP_848605.1), and non-human RIPK1 C-termini (cat, XP_023109490.2; pig, XP_003128209.1; chicken, NP_989733.3; lizard, XP_003224434.1; lamprey, XP_032813622.1; zebra mussel, XP_052232845.1) and RIPK3 (cat, XP_003987615.3; pig, XP_001927459.3; lizard, XP_003223896.2; turtle, XP_037771912.1; lamprey, XP_032816533.1) homologues were ordered from Twist Biosciences (San Francisco, CA, USA) and were cloned along with RHIM mutants targeting the core tetrad into the pcDNA5/FRT/TO backbone (Invitrogen, Carlsbad, CA) with an N-terminal V5 tag and linker using Gibson Assembly (New England Biolabs, Ipswich, MA). Kinase mutants for human proteins RIPK1 (D138N), RIPK2 (D146N), RIPK3 (D142N), RIPK4 (D143N), and RIPK5 (D145N), and RHIM mutants for RIPK1 (IQIG542AAAA) and RIPK3 (VQVG461AAAA) were generated using Gibson Assembly.

Cell culture and transient transfections

HEK293T cells (obtained from ATCC) and the generated HEK293T RIPK1 KO cells were maintained at a low passage number to maintain less than one year since purchase, acquisition, or generation. Both cell lines were grown in complete media containing DMEM (Gibco, Carlsbad, CA), 10% FBS (Peak Serum, Wellington, CO), and 1% penicillin/streptomycin (Gibco, Carlsbad, CA). A day prior to transfection, HEK293T and RIPK1 KO cells were seeded into 24-well plates with 500 μ L complete media or 96-well plates with 80 μ L complete media. Cells were transiently transfected with 500 ng total DNA and 1.5 μ L of TransIT-X2 (Mirus Bio, Madison, WI) in 100 μ L Opti-MEM (Gibco, Carlsbad,

CA) for a 24-well plate or 100 ng DNA and 0.3 μ L TransIT-X2 in 10 μ L Opti-MEM for a 96-well plate. DNA and TransIT-X2 were incubated at room temperature for 25-30 minutes, then added dropwise to the appropriate well. Cells were harvested or analyzed at 18-22 hours post-transfection.

Generation of knockout cell lines

RIPK1 knockout HEK293T cells were generated using CRISPR/Cas9 as previously described [53, 63]. Briefly, plasmids were generated order to produce lentivirus-like particles containing the CRISPR/Cas9 machinery and guide RNA targeting exon 5 (ENSE00003586162) of RIPK1. The protocol for the molecular cloning of this plasmid was adapted from Feng Zhang (Sanjana et al., 2014) using the transfer plasmid pLB-Cas9 (gift from Feng Zhang, Addgene plasmid # 52962). We designed the gRNA target sequence using the web tool CHOPCHOP (Labun et al., 2016), available at <https://chopchop.cbu.uib.no/>, and synthesized oligonucleotides from Integrated DNA Technologies (San Diego, CA). The synthesized oligonucleotide pair was phosphorylated and annealed using T4 Polynucleotide Kinase (NEB M0201S) with T4 Ligation Buffer (NEB). Duplexed oligonucleotides were ligated into dephosphorylated and BsmBI-digested pLB-Cas9 using the Quick Ligase kit (NEB M2200S) to generate transfer plasmid with RIPK1 guide sequence. To generate lentivirus-like particles, this transfer plasmid was transfected alongside two packaging plasmids, pMD2.G (gift from Didier Trono, Addgene plasmid # 12259) and psPAX2 (gift from Didier Trono, Addgene plasmid # 12260), into HEK293T cells with a 1:1:1 stoichiometry. Forty-eight hours post-transfection supernatant was harvested and syringe-filtered (0.45 μ m). Supernatant containing sgRNA-encoding lentivirus-like particles was then used to transduce HEK293T cells. Transduced cells were cultured in growth media for 48 hours, then cultured in growth media supplemented with 1 μ g/ml puromycin for 72 hours. Using limiting dilution in 96-well plates, the monoclonal cell line was then obtained. RIPK1 knockout was confirmed by Sanger sequencing and by western blot using an α -RIPK1 antibody.

NF- κ B and IFN luciferase reporter assays

To quantify NF- κ B and IFN activation, we used the Dual-Glo Luciferase Assay System E2920 (Promega, USA). WT or RIPK1 KO HEK293T cells were seeded in a white 96-well plate and transfected with firefly luciferase fused to either the NF- κ B response element (pGL4.32, Promega) or the human IFN-beta promoter (IFN-Beta-pGL3, Addgene #102597), renilla luciferase fused to herpes simplex virus thymidine kinase promoter (pTK-Renilla, Thermo Scientific), and the indicated RIPK. Firefly luciferase was used as the primary reporter with renilla luciferase being a normalization control. Eighteen to 24 hours after transfection, Dual-Glo Luciferase Assay Reagent was added to each transfected well as well as 3-6 untransfected wells to serve as a negative control. Following a 10-minute incubation, the firefly luciferase signal was measured using a BioTek Cytation imaging reader with Gen5 software (Agilent Technologies, San Diego, CA). Then, Dual-Glo Stop & Glo reagent was added to each well. Following a 10-minute incubation, the renilla luciferase signal was measured. The background luminescence signal from the buffer-treated untransfected conditions was subtracted from other conditions, firefly values were normalized to renilla values, and all samples were normalized to the empty vector control condition. Values are reported as NF- κ B activity (firefly/renilla).

Cell death assay

Cell viability was measured using the ReadyProbes Cell Viability Imaging Kit, Blue/Green (ThermoFisher Scientific). HEK293T cells were seeded in 80 μ L of DMEM+/+ (See "Cell culture and transient transfections") in a clear 96-well plate and transfected with the indicated plasmids or a vector control. Eighteen hours post-transfection, cells were stained using NucBlue Live reagent, which stains all nuclei, and NucGreen Dead reagent, which stains only dead cells. A 2x concentrated mix (4 drops/mL of each dye) was made and added to wells to 1x concentration. Cells were incubated for 20 minutes at room temperature. Fluorescence for NucBlue (DAPI, excitation 377/20, emission 447/20) and NucGreen (GFP, excitation 469/20, emission 525/20) were read using a

BioTek Cytation imaging reader with Gen5 software (Agilent Technologies, San Diego, CA). Unstained wells were read and used as controls for background fluorescence. GFP values were normalized to DAPI values, and all samples were normalized to the vector control condition. Values are reported as Cytotoxicity (GFP/DAPI).

Immunoblotting and antibodies

Eighteen to 22 hours post-transfection, cells were washed with 1X PBS and lysed with boiling 1x NuPAGE LDS sample buffer containing 5% β -mercaptoethanol at room temperature for 5 minutes and then at 98°C for 7-10 minutes. Lysates were separated by SDS-PAGE (4-15% Bis-tris gel; Life Technologies, San Diego, CA) with 1X MOPS buffer (Life Technologies, San Diego, CA). Proteins were transferred onto a nitrocellulose membrane (Life Technologies, San Diego, CA) and blocked with PBS-T containing 5% bovine serum albumin (BSA) (Spectrum, New Brunswick, NJ). Membranes were incubated with the indicated rabbit primary antibodies diluted with 5% BSA and PBS-T at 1:1000 overnight at 4°C (α -V5 clone D3H8Q, α -GAPDH clone 14C10, α -RIPK1 clone E8S7U XP, α -RIPK2 clone D10B11, α -RIPK3 clone E1Z1D, α -RIPK4 #12636; Cell Signaling Technology, Danvers, MA). Membranes were rinsed three times with PBS-T, incubated with HRP-conjugated rabbit secondary antibody diluted at 1:10,000 with 5% BSA and PBS-T for 30 minutes at room temperature, and developed with SuperSignal West Pico PLUS Chemiluminescent Substrate (Thermo Fisher Scientific, Carlsbad, CA).

Statistical analysis

Statistical analyses were completed using GraphPad Prism 10 software. Tests were performed as indicated. Error bars were calculated using SEM.

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Additional files

Supplemental files [↗](#)

Supplemental figures [↗](#)

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Reviewer #2 (Public review):

Summary:

By combining bioinformatical and experimental approaches, the authors address the question why several vertebrate lineages lack specific genes of the necroptosis pathway, or those that regulate the interplay between apoptosis and necroptosis. The lack of such genes was already known from previous publications, but the current manuscript provides a more in-depth analysis and also uses experiments in human cells to address the question of functionality of the remaining genes and pathways. A particular focus is placed on RIPK3/RIPK1 and their dual roles in inducing NFkB and/or necroptosis.

Strengths:

The well documented bioinformatical analyses provide a comprehensive data basis of the presence/absence of RIP-kinases, other RHIM proteins, apoptosis signaling proteins (FADD, CASP8, CASP10) and some other genes involved in these pathway. Several of these genes are known to be missing in certain animal lineages, which raises the question why their canonical binding partners are present in these species. By expressing several such proteins (both wildtype and mutants destroying particular interaction regions) in human cells, the authors succeed in establishing a general role of RIPK3 and RIPK1 in NFkB activation. This function appears to be better conserved and more universal than the necroptotic function of the RHIM proteins. The authors also scrutinize the importance of the kinase function and RHIM integrity for these separate functionalities.

Weaknesses:

A weakness of the presented study is the experimental restriction to human HEK293 cells. There are several situations where the functionality of proteins from distant organisms (like lampreys or even mussels) in human cells is not necessarily indicative of their function in native context. In some cases, these problems are addressed by co-expressing potential interaction partners, but not all of these experiments are really informative. However, I agree with the authors that it is not possible to perform all the experiments in native cells, and that comparing all proteins in the same (human) cell type allows for a better comparison.

The conclusions drawn by the authors are supported by convincing evidence. I have no doubts that this study will be very useful for future studies addressing the evolution of necroptosis and its regulation by NFkB and apoptosis.

Reviewer #3 (Public review):

In this study, the authors employ both computational and experimental methods to reveal functional conservation of RIP family kinases and associated proteins in animals, with particular focus on mammals and other major groups of vertebrates. The bioinformatic part of the work involves genomic data from diverse animal groups, providing insightful data on loss and duplications patterns for RIP and other necroptosis-related genes, and positive selection signals for RIPK1/3 genes in certain mammalian clades. These findings are then extensively used for selecting species and RHIM tetrad candidates for further experiments, in which the authors demonstrate different modes of functional conservation for RIPK proteins in necroptosis and NF- κ B signaling across vertebrate species.

As an only major drawback, I would mention several important findings which the authors make in the course of their research but do not pursue further in the experimental part of the paper. These include:

- An additional copy for RIPK2 (RIPK2B) found in monotremes and non-mammalian vertebrates and its functions;
- The entire diversity of RHIM functional tetrad variants; of particular interest here are IQFG and IQLG tetrads specific for bats, which are known to harbor human-affecting viruses and were demonstrated to have their RIPK1/3 genes under positive selection in this study;
- Functions and involvement of RIPK3 protein in NF- κ B pathway in lampreys;
- The mode of NF- κ B activation in non-mammalian species retaining ZBP1 copies.

Further elucidation of some or all of these points in the experimental part would facilitate conceptualizing the paper's numerous findings, which otherwise might appear insufficiently scrutinized. On the other hand, I agree that at least some of them require separate studies to be elucidated in. Given the importance of the results presented in this paper, I believe these points will be further addressed in future works.

<https://doi.org/10.7554/eLife.102301.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

The manuscript titled "Evolutionary and Functional Analyses Reveal a Role for the RHIM in Tuning RIPK3 Activity Across Vertebrates" by Fay et al. explores the function of RIPK gene family members across a wide range of vertebrate and invertebrate species through a combination of phylogenomics and functional studies. By overexpressing these genes in human cell lines, the authors examine their capacity to activate NF- κ B and induce cell death. The methods employed are appropriate, with a thorough analysis of gene loss, positive selection, and functionality. While the study is well-executed and comprehensive, its broader relevance remains limited, appealing mainly to specialists in this specific field of research. It misses the opportunity to extract broader insights that could extend the understanding of these genes beyond evolutionary conservation, particularly by employing evolutionary approaches to explore more generalizable functions.

Major comments:

The main issue I encounter is distinguishing between what is novel in this study and what has been previously demonstrated. What new insights have been gained here that are of broader relevance? The discussion, which would be a good place to do so, is very speculative and has little to do with the actual results. Throughout the manuscript, there is little explanation of the study's importance beyond the fact that it was possible to conduct it. Is the evolutionary analysis being used to advance our understanding of gene function, or is the focus merely on how these genes behave across different species? The former would be exciting, while the latter feels less impactful.

We thank the reviewer for the positive feedback. With regard to the major comment, we have now made changes throughout the revised manuscript to highlight the novel insights that emerge from our work, as well as the importance of using evolutionary and functional analyses to understand gene function.

Reviewer #2 (Public review):

Summary:

By combining bioinformatical and experimental approaches, the authors address the question of why several vertebrate lineages lack specific genes of the necroptosis pathway or those that regulate the interplay between apoptosis and necroptosis. The lack of such genes was already known from previous publications, but the current manuscript provides a more in-depth analysis and also uses experiments in human cells to address the question of the functionality of the remaining genes and pathways. A particular focus is placed on RIPK3/RIPK1 and their dual roles in inducing NFκB and/or necroptosis.

Strengths:

The well-documented bioinformatical analyses provide a comprehensive data basis of the presence/absence of RIP-kinases, other RHIM proteins, apoptosis signaling proteins (FADD, CASP8, CASP10), and some other genes involved in these pathways. Several of these genes are known to be missing in certain animal lineages, which raises the question of why their canonical binding partners are present in these species. By expressing several such proteins (both wildtype and mutants destroying particular interaction regions) in human cells, the authors succeed in establishing a general role of RIPK3 and RIPK1 in NFκB activation. This function appears to be better conserved and more universal than the necroptotic function of the RHIM proteins. The authors also scrutinize the importance of the kinase function and RHIM integrity for these separate functionalities.

Weaknesses:

A major weakness of the presented study is the experimental restriction to human HEK293 cells. There are several situations where the functionality of proteins from distant organisms (like lampreys or even mussels) in human cells is not necessarily indicative of their function in the native context. In some cases, these problems are addressed by co-expressing potential interaction partners, but not all of these experiments are really informative.

A second weakness is that the manuscript addresses some interesting effects only superficially. By using host cells that are deleted for certain signaling components, a more focussed hypothesis could have been tested.

Thus, while the aim of the study is mostly met, it could have been a bit more ambitious. The limited conclusions drawn by the authors are supported by convincing evidence. I have no doubts that this study will be very useful for future studies addressing the evolution of necroptosis and its regulation by NFkB and apoptosis.

We thank the reviewer for the positive feedback. We agree that our study is limited by using HEK293 cells. However, we do not have appropriate cell lines for all species analyzed and therefore wished to use a single system to test all effects. As the reviewer points out, we do co-express when possible, and are careful in the manuscript to not overextend our conclusions. We, like the reviewer, believe that many of the intriguingly findings in this study, which was intended to cover a broad range of species, will be useful for more in-depth studies in a given species.

Reviewer #3 (Public review):

This important study provides insights into the functional diversification of RIP family kinase proteins in vertebrate animals. The provided results, which combine bioinformatic and experimental analyses, will be of interest to specialists in both immunology and evolutionary biology. However, the computational part of the methodology is insufficiently covered in the paper and the experimental results would benefit from including data for additional species.

We thank the reviewer for the positive feedback. As described below, we have now addressed the concerns about the description of the computational methods.

(1) In the Methods section concerning gene loss analysis, the authors refer to the 'Phylogenetic analysis' section for details of RIPK sequence acquisition and alignment procedure. This section is missing from the manuscript as provided. In its absence, it is hard for the reviewer to provide relevant comments on gene presence/absence analysis.

We have expanded the gene loss analysis methods to be more comprehensive.

(2) In the same section, the authors state that gene sequences were filtered and grouped based on the initial gene tree pattern (lines 448-449). How exactly did the authors filter the non-RIP kinases and other irrelevant homologs from the gene trees? Did they consider the reciprocal best (BLAST) hit approach or similar approaches for orthology inference? Did they also encounter potential pseudogenes of genes marked as missing in Figure 1C? Will the gene trees mentioned be available as supplementary files?

We have expanded the gene loss analysis methods to be more comprehensive.

(3) The authors state the presence of additional RIPK2 paralog in non-therian vertebrates.

The ramifications of this paralog loss in therians are not discussed in the text, although RIPK2 is also involved in NF-kB activation. In addition, the RIPK2B gene loss pattern is shunned from Figure 1C to Supplementary Figure 4, despite posing comparable interest to the reader.

We are also intrigued by the RIPK2/RIPK2B data and felt it important to include our findings here, however we do not have functional data for RIPK2B at this point and feel it is better suited for a separate study. We therefore focused both the title and the main figures on RIPK3, for which we have functional data.

(4) The authors present evidence for (repeated) positive selection in both RIPK1 and RIPK3 in bats; however, neither bat RIPK1/3 orthologs nor bat-specific RHIM tetrad variants (IQFG, IQLG) are considered in the experimental part of the work.

We included a tetrad variant (VQFG) that is found in bats and multiple other species. We wanted to test a wide range of variant amino acids, so testing both IQFG (found only in bats) and VQFG (found in bats and multiple other diverse species) was not of high importance.

(5) The authors present gene presence/absence patterns for zebra mussels as an outgroup of vertebrate species analyzed. From the evolutionary perspective, adding results for a closer invertebrate group, such as lancelets, tunicates, or echinoderms, would be beneficial for reconstructing the evolutionary progression of RIPK-mediated immune functions in animals.

In our initial analyses, we searched for RIPK-like proteins in cnidarians, arthropods, nematodes, amoeba, and spiralia, with only spiralia species containing proteins with substantial homology to vertebrate RIPK1 proteins, as defined by a homologous N-terminal kinase domain and C-terminal RHIM and death domain. We have expanded this analysis to include lancelets, tunicates, and echinoderms and found several lancelet species with RIPK1 like proteins. These data have been added to the manuscript.

(6) In the broader sense, the list of non-mammalian species included in the study is not explained or substantiated in the text. What was the rationale behind selecting lizards, turtles, and lampreys for experimental assays? Why was turtle RIPK3 but not turtle RIPK1CT protein used for functional tests? Which results do the authors expect to observe if amphibian or teleost RIPK1/3 are included in the analysis, especially those with divergent tetrad variants?

We have added additional text to define our rationale for selecting which species were tested.

(7) For lamprey RIPK3, the observed NF- κ B activity levels still remain lower than those of mammalian and reptilian orthologs even after catalytic tetrad modification. In the same way, switching human RIPK3 catalytic tetrad to that of lamprey does not result in NF- κ B activation. What are the potential reasons for the observed difference? Does it mean that lamprey's RIPK3 functions in NF- κ B activation are, at least partially, delegated to RIPK1?

The function of lamprey RIPK3 is intriguing, albeit unknown. The reduced activation in human cells may be due to an incompatibility between lamprey RIPK3 and human NF- κ B machinery, or it may not function in NF- κ B at all. Considering that lamprey do not have other components of the known mammalian necroptosis pathway, it is unclear what function RIPK3 would serve in these species. It is possible lamprey may have a necroptosis pathway that is RIPK3-dependent but distinct from the mammalian pathway. It is an interesting question for future study.

(8) In lines 386-388, the authors state that 'only non-mammalian RIPK1CT proteins required the RHIM for maximal NF- κ B activation', which is corroborated by results in Figure 4B. The authors further associate this finding with a lack of ZBP1 in the respective species (lines 388-389). However, non-squamate reptiles seem to retain ZBP1, as suggested by

Supplementary Table 1. Given that, do the authors expect to observe RHIM-independent (maximal) NF- κ B activation in turtles and crocodilians or respective RIPK1CT-transfected cells?

While turtles and crocodiles do retain ZBP1, it is still unclear if they are able to activate ZBP1/RIPK3/MLKL-dependent necroptosis similar to mammals, especially given the divergence in the turtle ZBP1 RHIMs seen in Figure 4C. Future studies will be needed to further test our hypotheses and to continue to characterize innate immune function and evolution across a range of vertebrate species.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

Minor comments:

(1) The title is somewhat restrictive, as it only mentions RIPK3, despite the manuscript covering a broader range of RIPKs and associated proteins.

We agree that a title that encompasses both the breadth of our study and the depth with which we analyzed RIPK3 would be ideal. However, we were unable to come up with a succinct title that conveyed both points appropriately, so opted for one that focused on our RIPK3 insights.

(2) Several supplementary figures contain valuable information that could be incorporated into the main figures for greater clarity and emphasis.

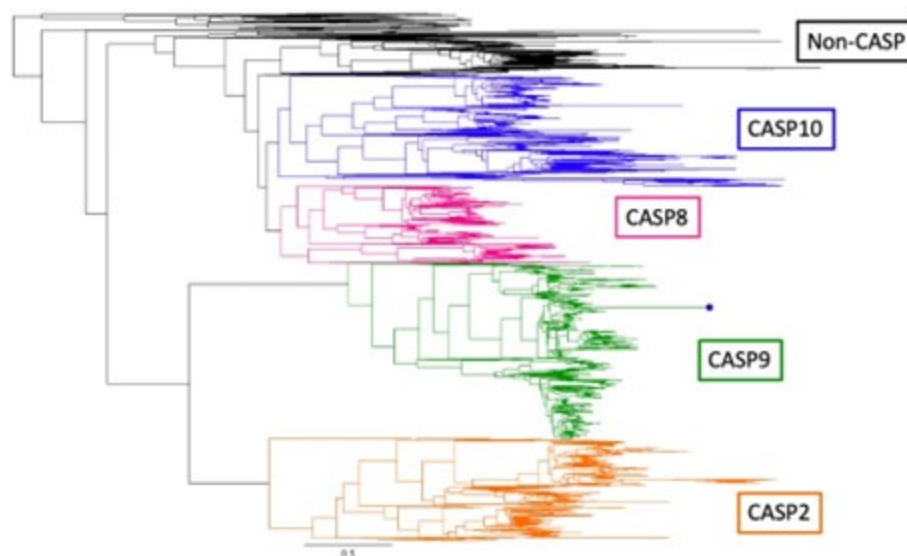
We agree that many interesting pieces of data are in the supplement. We felt it was important to include those data in the manuscript, but also wanted to keep the main manuscript figures as focused as possible.

Reviewer #2 (Recommendations for the authors):

(1) I do not fully agree with the claim that caspase-8 is absent from fish. I briefly repeated this part of the analysis and found several fish proteins that cluster with caspase-8 rather than caspase-10 or cFLIP. From the method section, it does not really become clear how the Casp8/Casp10/cFLIP decision was made, and particularly, how cases were addressed where Genew predate the caspase-8/caspase-10 split. To name just a few examples, the authors might check uniprot:A0A444UA91, W5MXS4, or A0A8X8BKJ8 for being fish Caspase-8 candidates.

We thank the reviewer for their critical analysis. CASP8 and CASP10 are very similar proteins in humans. We are distinguishing between the two based on vertebrate phylogeny with outgroup proteins (CASP2, CASP9, and CFLAR, see tree in Author response image 1 below) to help define the CASP8/CASP10 clade. Once we isolate CASP8/10, we build an additional tree to distinguish CASP8 and CASP10. Using this method, all fish CASP8/10-like proteins cluster with the mammalian CASP10 clade rather than the CASP8 clade, despite many fish proteins being annotated as CASP8 or CASP8-like. We do acknowledge that, because of the similarities between CASP8 and CASP10, there are likely proteins that can fall in either clade depending on which outgroups are included. To this end, we have updated our gene loss figure to only denote whether a species has no CASP8/10, a single CASP8/10 protein, or both CASP8 and CASP10. We have also updated our methods to better define how we completed our analyses.

Author response image 1.

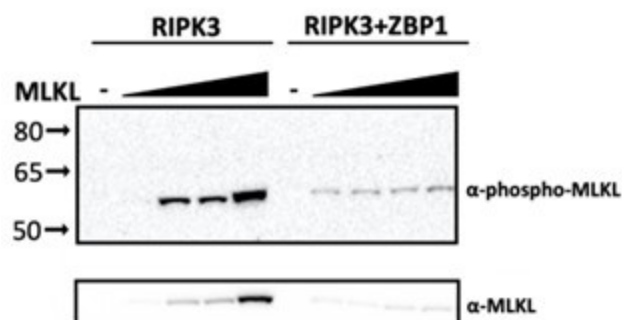


(2) While analyzing which RIPK3 protein causes cell death (lines 188ff), the underlying assumption is that the heterologous RIPK3 proteins can interact with human MLKL and activate it by phosphorylation. No attempts are being made to check if MLKL actually gets phosphorylated, and this issue is also not discussed. In Figure 2C, cell death is either measured by RIPK3 overexpression alone or by the additional overexpression of ZBP1 and MLKL. However, it is not shown if in all cases all the transfected proteins are expressed at a comparable level, or if the observed cell death might be caused by MLKL/ZBP1 overexpression alone.

Cell death is dependent on expression of ZBP1, MLKL, and RIPK3, as shown in

Supplementary Figure 6. We have attempted to detect phospho-MLKL via western blot. However, in these overexpression assays, we are able to detect phospho-MLKL in the presence of RIPK3 and MLKL alone, independent of activation of cell death. In fact, we see reduced phospho-MLKL and reduced expression of MLKL overall when ZBP1, MLKL, and RIPK3 are added, presumably due to cell death induced in these conditions (see blot in Author response image 2 below). We therefore felt these data were of limited use here.

Author response image 2.



(3) The manuscript describes a well-documented bioinformatical analysis and acknowledges the body of earlier published work on necroptosis evolution and associated gene losses. However, when discussing the RHIM-related aspects, the authors do not mention previous publications on RHIM conservation in invertebrates and even fungal proteins such as Het-S. They also fail to mention/discuss the amyloid-forming properties of RHIMs, which I consider crucial for understanding the function of RHIM-containing proteins.

We thank the reviewer for their insight. We have added additional points on both RHIM conservation and amyloid formation.

(4) Related to the above issue: In lines 226ff, the induction of NFκB by RIPK3 overexpression is described. While RIPK3 from other mammals requires endogenous (human) RIPK1 to be present, lizard and turtle RIPK3 do not require human RIPK1 but *do* require functional RHIMs. It is not checked (or at least discussed) if RHIM amyloid formation is required, nor if the RHIM of the heterologous RIPK3 might act through interaction with endogenous (human) RIPK3.

We and others (PMID: 29073079) did not detect RIPK3 protein in HEK293T cells. This, combined with the requirement for exogenous RIPK3 to activate cell death, indicate that endogenous RIPK3 is not contributing to these assays.

(5) In lines 275ff, the authors observe that RIPK1s from other mammalian species do not require the RHIM for NFκB activation, while RIPK1 from non-mammalian species do require the RHIM. I wonder why the (in my opinion) most obvious explanation is not addressed: Maybe the mammalian RIPK1 proteins are similar enough to the human one so that they can signal on their own, while the more distant RIPK1 cannot and thus require human RIPK1 (associated via RHIMs) for NFκB activation? Since the authors used RIPK1-deficient cells in previous experiments, wouldn't it make sense to test them here, too?

It is intriguing that the more diverged RIPK1 species require the RHIM for NF-κB signaling. In Supplementary Figure 12, we do test the mammalian and non-mammalian proteins in RIPK1 KO cells and all proteins are able to activate NF-κB. So while nonmammalian RIPK1 signaling is dependent on the RHIM, it is independent of endogenous RIPK1.

Minor comments:

(1) In the legend of Figure 1, there is a typo "heat amp".

This typo has now been corrected.

(2) In Figure 3A, the term "FUBAR" is not explained at all.

FUBAR has now been defined in the methods section.

Reviewer #3 (Recommendations for the authors):

A few typos and graph inconsistencies have been encountered in the course of the manuscript, e.g.:

(1) Line 168: 'heat amp' -> 'heat map'.

(2) Lines 290-291: 'known mediate' -> 'known to mediate' (?)

We thank the reviewer for catching these mistakes. They have been corrected.

(3) Supplementary Figure 12: Are human RIPK1 results presented in both 'mammalian' and 'non-mammalian' parts of the figure? If so, why do human data differ between the graphs?

Mammalian and non-mammalian data were collected in separate experiments with human RIPK1 used as a control for both. The human data shown in the two graphs represent two separate experiments.

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